

NEUROSCI 366

Lecture 17

视觉网络——Cascades, Complex Cells, and Recurrence

Source-aligned · Static · Searchable · Printable

Source files

- [Lecture17-VisualNetworks.pdf](#)

Production version: 2026-08-22

Reader profile: neuroscience undergraduate education; mathematics scaffolded from high-school algebra/trigonometry to the formal computational-neuroscience level.

This companion preserves source-page order, distinguishes source material from necessary scaffolding and extensions, and places hints/answers after the lesson. It is a static PDF: no forms, JavaScript, hidden-answer widgets, passwords, or DRM.

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1 How to use this companion

1. 先用 5-minute diagnostic 暴露前置缺口；不要立即翻答案。
2. 按原笔记页序阅读 source-page units：先看原页，再读 clean reconstruction 与补全逻辑。
3. 遇到公式按“问题 → 符号/shape/units → assumptions → 一行一步推导 → intuition → biological meaning → sanity check → failure”阅读。
4. 完成 Cumulative Knowledge Check；只在需要时依次打开 Hint 1、2、3。
5. 至少隔一个 page break 后再核对 Answer Key，并记录错误类型：concept、symbol、algebra、assumption、correlation/causation。

Source hierarchy. 原笔记是内容权威；本书中的“【必要前置】/【补全推导】/【跨课连接】/【延伸】”是为了补齐数学或迁移，不替代源材料。无法确认的内容必须进入 Uncertainty Log。

2 Learning objectives and prerequisite dependency map

- 定义并区分：LNP cascade。
- 从第一原则推导：simple-cell pooling。
- 解释几何/计算直觉：complex-cell STA=0。
- 连接生物学解释：quadrature energy。
- 诊断 assumptions 与 failure modes：task-optimized models。
- 把概念迁移到新神经科学问题：E/I recurrence。

Dependency map

LNP cascade → simple-cell pooling → complex-cell STA=0 → quadrature energy → task-optimized models → E/I recurrence

Core question

多层 LNP cascades 与 quadratic energy model 如何产生 selectivity/invariance, recurrent E/I dynamics 又如何解释 surround suppression?

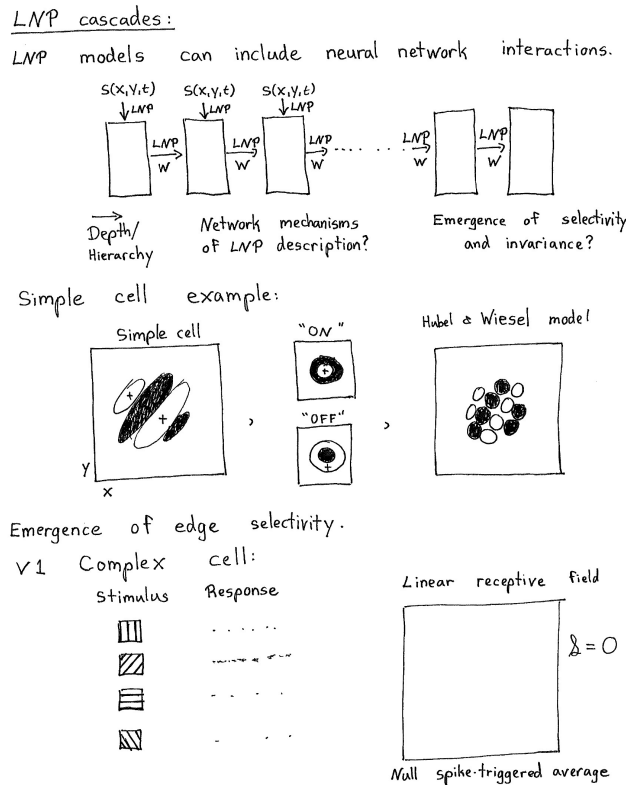
3 Five-minute prerequisite diagnostic

1. Hubel-Wiesel simple-cell construction 的核心？（卡住时先看 Source-page unit 1，再看补全推导。）
2. complex cell STA 为什么可为零？（卡住时先看 Source-page unit 2，再看补全推导。）
3. energy model 需要哪两类 filters？（卡住时先看 Source-page unit 3，再看补全推导。）
4. 证明 phase invariance 的 identity？（卡住时先看 Source-page unit 4，再看补全推导。）
5. task accuracy 高是否保证 neural predictivity 高？（卡住时先看 Source-page unit 5，再看补全推导。）

Scoring guide: 4–5 confident answers → proceed; 2–3 → use the prerequisite labels; 0–1 → read the formula/notation sheet once before the source-aligned lesson.

4 Source-aligned lesson / 按原笔记页序

4.1 Lecture17-VisualNetworks.pdf • p. 1



Original-note anchor. [Source: Lecture17-VisualNetworks.pdf, p. 1]

Clean reconstruction. Page 1: LNP cascades, Hubel-Wiesel simple-cell construction, complex-cell zero STA.

What the note is saying. 本页的中心对象是 LNP cascade 与 simple-cell pooling。原笔记将它们放进以下链条：LNP cascades、Hubel-Wiesel simple-cell construction、complex-cell zero STA。

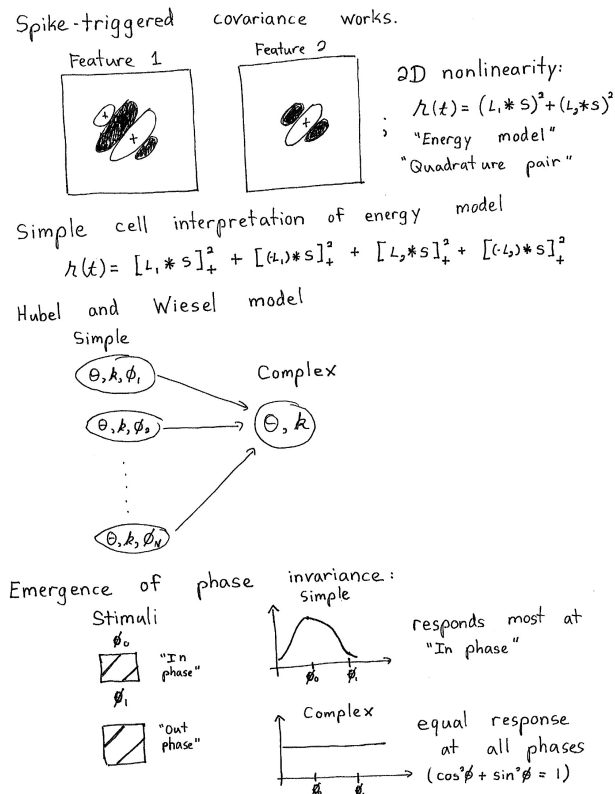
Why it follows. 重构 Hubel-Wiesel simple-cell pooling 与 complex-cell phase invariance；说明 complex cell 为什么 linear STA 可为零。其逻辑后果是：解释 LNP cascades 如何产生 hierarchy、selectivity 与 invariance；区分 descriptive cascade 与 biological mechanism。后面的正式推导会逐项写出 assumptions、shape/units、limiting cases 与 failure conditions。

Figure reading. 先识别图中对象、箭头、参数和坐标系，再问它表达的是定义、推导、模型预测还是经验观察；示意图不提供未标注的定量精度。

Stop & Predict

Hubel-Wiesel simple-cell construction 的核心？

4.2 Lecture17-VisualNetworks.pdf • p. 2



Original-note anchor. [Source: Lecture17-VisualNetworks.pdf, p. 2]

Clean reconstruction. Page 2: STC features, energy model, quadrature pair, phase invariance.

What the note is saying. 本页的中心对象是 complex-cell STA=0 与 quadrature energy。原笔记将它们放进以下链条：STC features、energy model、quadrature pair、phase invariance。

Why it follows. 逐步解释 energy model、quadrature filters、square-and-sum 与 phase invariance；连接 STC multiple features。其逻辑后果是：重构 Hubel-Wiesel simple-cell pooling 与 complex-cell phase invariance；说明 complex cell 为什么 linear STA 可为零。后面的正式推导会逐项写出 assumptions、shape/units、limiting cases 与 failure conditions。

Figure reading. 读图时先标出两条 nullcline/坐标轴，再判断向量场方向、交点稳定性与轨迹穿越顺序；轨迹本身不等于因果机制。

Stop & Predict

complex cell STA 为什么可为零？

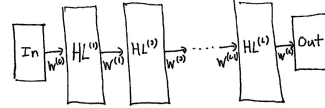
4.3 Lecture17-VisualNetworks.pdf • p. 3

Performance-optimized models

IT is deep in the hierarchy, highly selective for objects, and highly invariant to details.

How can we model such complex neural responses?

Step 1: Build a model that can classify objects well



$$\{\hat{W}^{(1)}, \hat{W}^{(2)}, \dots, \hat{W}^{(L)}\} = \arg \min_{\{W^{(1)}, \dots, W^{(L)}\}} \mathcal{L}(W^{(1)}, \dots, W^{(L)})$$

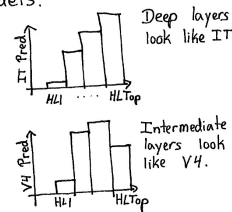
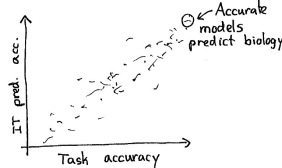
Optimize with machine learning Quantifies object recognition performance (classification)

Step 2: Model biological neurons with artificial ones

$$\hat{h}_i = \sum_{j=1}^N \underbrace{\hat{\phi}_{ij}}_{\text{Weighting coefficients}} \underbrace{Z_j}_{\text{Artificial}}; \quad \hat{\phi} = \arg \min_{\phi} \sum_{i \in M} (\underbrace{h_{iM} - \hat{h}_{iM}}_{\text{Mean squared error}})^2 / P$$

Bio Artificial Mean squared error

Paradigm pioneered by Yamins and DiCarlo and produces state of the art models.



Original-note anchor. [Source: Lecture17-VisualNetworks.pdf, p. 3]

Clean reconstruction. Page 3: performance-optimized object-classification models 与 neural predictivity.

What the note is saying. 本页的中心对象是 task-optimized models 与 E/I recurrence。原笔记将它们放进以下链条：performance-optimized object-classification models 与 neural predictivity。

Why it follows. 解释 performance-optimized deep network 的两阶段：先优化任务，再拟合/比较 biological responses；区分 task accuracy 与 neural predictivity。其逻辑后果是：逐步解释 energy model、quadrature filters、square-and-sum 与 phase invariance；连接 STC multiple features。后面的正式推导会逐项写出 assumptions、shape/units、limiting cases 与 failure conditions。

Figure reading. 先识别图中对象、箭头、参数和坐标系，再问它表达的是定义、推导、模型预测还是经验观察；示意图不提供未标注的定量精度。

Stop & Predict

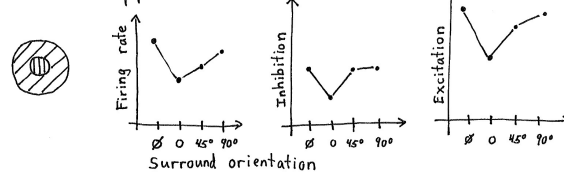
energy model 需要哪两类 filters?

4.4 Lecture17-VisualNetworks.pdf • p. 4

Recurrence in vision:

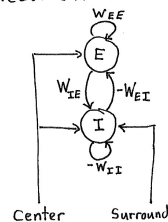
We've so far emphasized the power of deep, feedforward processing. Visual neurons also interconnect within a region, and this has important implications for neural coding.

Surround suppression:



The surround decreases the firing rate, which naively suggests that the surround inhibits the cell. In 2009, David Ferster's lab at Northwestern surprisingly found that the surround decreases the inhibition. The firing rate goes down because excitation decreases more.

Recurrent neural network model:



$$\tau_E \frac{d\tau_E}{dt} = -\tau_E + \sum_E (W_{EE}\tau_E - W_{EI}\tau_I + I_E)$$

$$\tau_I \frac{d\tau_I}{dt} = -\tau_I + \sum_I (-W_{II}\tau_I + W_{IE}\tau_E + I_I)$$

Original-note anchor. [Source: Lecture17-VisualNetworks.pdf, p. 4]

Clean reconstruction. Page 4: recurrence, surround suppression, E/I rate-network model.

What the note is saying. 本页的中心对象是 nullclines/ISN 与 LNP cascade。原笔记将它们放进以下链条: recurrence、surround suppression、E/I rate-network model。

Why it follows. 分析 surround suppression 不必等于更多 inhibition: 可由 excitation decrease more than inhibition 产生。其逻辑后果是: 写出 E/I rate equations、fixed-point conditions 与 nullclines; 解释 surround input 如何移位 nullcline 并降低 E/I rates。后面的正式推导会逐项写出 assumptions、shape/units、limiting cases 与 failure conditions。

Figure reading. 逐条追踪箭头方向、兴奋/抑制符号、状态变量与外部输入; 结构图说明允许的依赖关系, 不自动证明生物实现。

Stop & Predict

证明 phase invariance 的 identity?

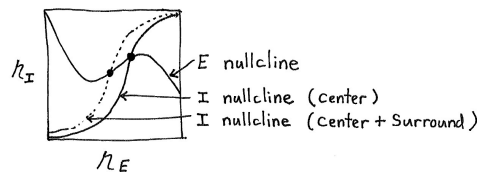
4.5 Lecture17-VisualNetworks.pdf • p. 5

The firing rates settle at values such that both derivatives equal zero.

$$\frac{dh_E}{dt} = 0 \Rightarrow h_E = \sum_E (W_{EE} h_E - W_{EI} h_I + I_E)$$

$$\frac{dh_I}{dt} = 0 \Rightarrow h_I = \sum_I (W_{IE} h_E - W_{II} h_I + I_I)$$

These define two curves in (h_E, h_I) space, termed nullclines, whose intersections determine possible steady states of the network. In the same 2009 paper, Ken Miller's lab proposed an explanation for the Ferster experimental result.



The surround input increases I_I , shifting the I nullcline upwards. Both h_E and h_I are decreased in the new steady state, thereby decreasing both excitation and inhibition to the excitatory cells.

The nullcline geometry above indicates an "inhibition stabilized network." We need to better understand neural network dynamics to appreciate the name.

Original-note anchor. [Source: Lecture17-VisualNetworks.pdf, p. 5]

Clean reconstruction. Page 5: steady-state nullclines, surround-induced shifts, inhibition-stabilized network.

What the note is saying. 本页的中心对象是 simple-cell pooling 与 complex-cell STA=0。原笔记将它们放进以下链条：steady-state nullclines、surround-induced shifts、inhibition-stabilized network。

Why it follows. 给出 inhibition-stabilized network 的定义、直觉和为何名称容易被误解。其逻辑后果是：解释 performance-optimized deep network 的两阶段：先优化任务，再拟合/比较 biological responses；区分 task accuracy 与 neural predictivity。后面的正式推导会逐项写出 assumptions、shape/units、limiting cases 与 failure conditions。

Figure reading. 读图时先标出两条 nullcline/坐标轴，再判断向量场方向、交点稳定性与轨迹穿越顺序；轨迹本身不等于因果机制。

Stop & Predict

task accuracy 高是否保证 neural predictivity 高？

5 补全推导与数学支架 / Derivations

5.1 Hubel–Wiesel pooling 与 simple-cell selectivity

【必要前置】 如何由 center-surround subunits 形成 oriented RF?

$$L = \sum_j w_j r_j^{RGC/LGN}, \quad h = [L]_+$$

1. 将空间上排成一列的 ON/OFF subunits 赋予合适正负 weights。
2. aligned edge 同时激活多个同号 subunits, linear sum 大。
3. orthogonal/错位 edge 正负贡献抵消。
4. rectification 产生 firing-rate selectivity。

Geometric/computational intuition. orientation selectivity 可由多个较小 RF 的几何排列涌现。
Biological interpretation / failure condition. 真实 V1 selectivity 还受 recurrence、conductance 与 adaptation 调节。

Micro-check

证明 phase invariance 的 identity?

5.2 quadrature energy model 与 phase invariance

【补全推导】 为什么两个相差 90° 的 filters square-and-sum 后不依 phase?

$$L_1 = A \cos \phi, \quad L_2 = A \sin \phi$$

$$E = L_1^2 + L_2^2 = A^2(\cos^2 \phi + \sin^2 \phi) = A^2$$

1. quadrature pair 具有相同 orientation/frequency, phase 差 $\pi/2$ 。
2. 同一 grating phase 在两个 filters 上投影成 cos 与 sin components。
3. 平方消除符号, 求和利用 trigonometric identity。
4. 输出保留 orientation/frequency energy, 却消除 phase dependence。

Geometric/computational intuition. complex cell 检测某 orientation 的局部能量, 而非精确亮暗 lobe alignment。

Biological interpretation / failure condition. 有限 filter set、nonideal quadrature 或 normalization 会留下 phase modulation。

Micro-check

task accuracy 高是否保证 neural predictivity 高?

5.3 performance-optimized model 与 neural predictivity

【跨页连接】 如何把 object-classification model 与 IT responses 比较?

$$\widehat{W} = \arg \min_W \mathcal{L}_{task}(W)$$

$$\widehat{\beta} = \arg \min_{\beta} \sum_{\mu, i} (r_{i\mu} - \beta_i^T z_{\mu}(W))^2$$

1. 第一阶段只优化 task performance, 得到 internal features z 。
2. 第二阶段用线性/简单 mapping 预测 biological neural responses。
3. task accuracy 与 neural predictivity 是不同 axes。
4. 好的 task model 可产生类似 hierarchy, 但不证明 brain 使用相同 training/data/mechanism。

Geometric/computational intuition. 先问 “解决任务会产生什么 representation”, 再问 “它是否像 brain”。

Biological interpretation / failure condition. 过度解释 correlation, 忽略 architecture、training set 与 recording noise。

Micro-check

model features 与 neural responses 常如何比较?

5.4 E/I nullclines 与 surround suppression

【模型边界】 surround input 如何让 E、I rates 同时下降？

$$\tau_E \dot{r}_E = -r_E + F_E(W_{EE}r_E - W_{EI}r_I + I_E)$$

$$\tau_I \dot{r}_I = -r_I + F_I(W_{IE}r_E - W_{II}r_I + I_I)$$

1. steady state 是两条 nullclines 的 intersection。
2. surround 可增加 inhibitory-population drive, 使 I-nullcline 移位。
3. 新 intersection 可有更低 r_E 和更低 r_I ; excitation decrease 更大使 pyramidal firing suppressed。
4. 若 E subnetwork 单独不稳定但由 feedback inhibition 稳定, 称 inhibition-stabilized network。

Geometric/computational intuition. “inhibition stabilizes” 不等于 steady-state inhibitory rate 必须上升。

Biological interpretation / failure condition. nullcline conclusion 依 nonlinear gains/weights; 不同 circuits 可给相同 phenomenology。

Micro-check

surround suppression 是否要求 inhibition 增加？

6 Cross-page synthesis / 跨页因果与数学链

feedforward cascades 可逐级构造 selectivity 与 invariance, 但 descriptive blocks 不自动指定 biological mechanism。complex cells 对 grating phase 近 invariant, 因而单一 linear STA 可为零; quadrature-pair square-and-sum 的能量 model 给出最小解释。surround suppression 也不能简单等于 “更多 inhibition”: recurrent E/I network 可使 excitation 和 inhibition 同时下降, 只是 excitation 降得更多; nullcline geometry 揭示 inhibition-stabilized regime。

Lecture 17 — key reconstruction (schematic)

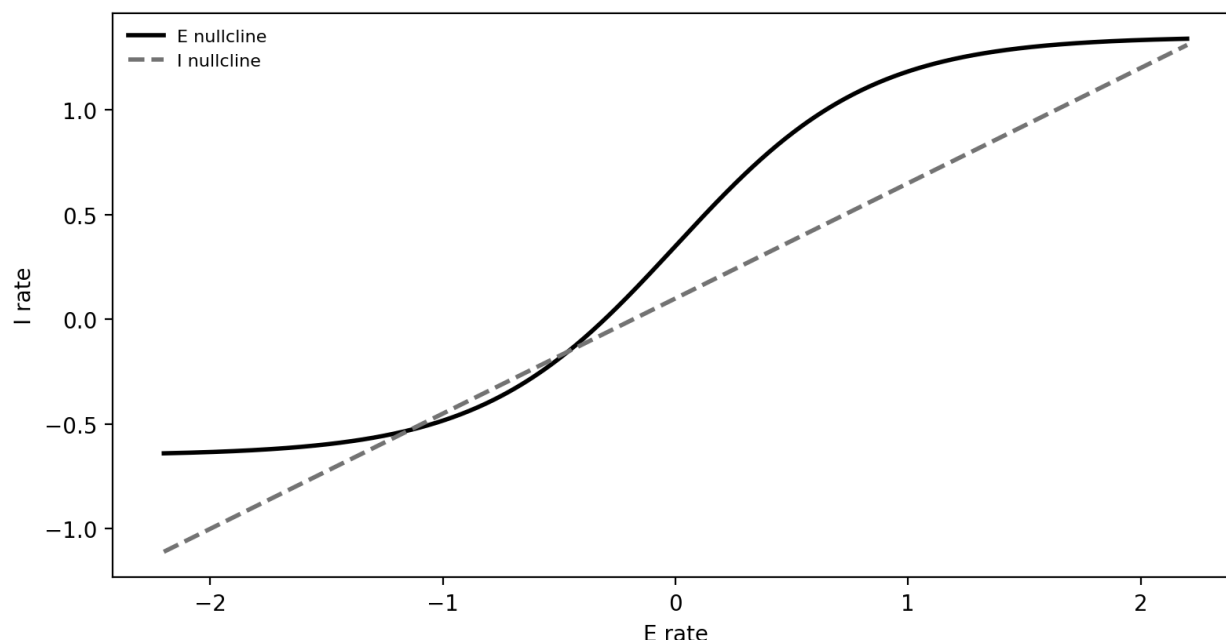


Figure note: schematic reconstruction; it encodes qualitative relations from the lesson and does not invent precise empirical data points.

7 Worked examples and transfer

7.1 STA=0 但 selectivity 强

【原课/核心例题】 Prompt. complex cell 对 0° grating 的所有 phases 都响应。

Worked solution. spike-triggered stimuli 含相反 bright/dark phases, linear average 抵消; STC 找到两个 quadrature features, energy output 保持高。

Check. zero first moment 不等于 zero higher-order dependence。

7.2 surround mechanism diagnosis

【跨课连接】 Prompt. 加 surround 后 excitatory conductance 与 inhibitory conductance 都下降, 前者下降更多。

Worked solution. net depolarizing drive 下降, firing suppression 仍可发生; 不能以 inhibition 未增加否定 recurrent inhibitory stabilization。

Check. 要比较 net currents/operating point, 而不只是单个 conductance direction。

Micro-check

Explain in your own words. 用不超过 120 字解释: 本课从 source page 1 到最后一页, 究竟把哪一个输入、状态、统计量或学习规则变成了可检验 prediction?

Self-reflection. 你最可能犯的是 concept、symbol、algebra、assumption 还是 causality error? 写出一个具体例子。

8 Formula and notation sheet

1. motion/orientation energy model

$$E = (K_1 * s)^2 + (K_2 * s)^2$$

Conditions / conventions: quadrature filters

2. phase-invariance identity

$$\cos^2 \phi + \sin^2 \phi = 1$$

Conditions / conventions: ideal pair

3. model-to-neuron linear mapping

$$\hat{r} = Bz$$

Conditions / conventions: features z fixed

4. E/I nullclines

$$\dot{r}_E = 0, \dot{r}_I = 0$$

Conditions / conventions: steady state

5. rate-network fixed point

$$r = F(Wr + I)$$

Conditions / conventions: chosen nonlinearities

6. ISN criterion intuition

E-only unstable + feedback I stable

Conditions / conventions: local effective gain matters

9 Bilingual glossary

中文	English / symbol	Operational meaning
级联	cascade	多个 processing stages 依次组合。
简单细胞	simple cell	对 orientation 且对 spatial phase 敏感的经典 V1 类型。
复杂细胞	complex cell	对 orientation 选择但对 phase 更 invariant 的经典 V1 类型。
正交相位对 能量模型	quadrature pair energy model	phase 相差 90° 的匹配 filters。 quadrature outputs square-and-sum 的 nonlinear model。
神经预测性 周围抑制	neural predictivity surround suppression	model features 对 neural responses 的预测程度。 刺激 RF surround 时 center response 减少。
零流线	nullcline	某 state derivative 为零的 phase-plane curve。
抑制稳定网络	inhibition-stabilized network, ISN	E subnetwork 单独不稳定、由 feedback inhibition 稳定。

10 Common traps, assumptions, and limitations

1. 把 descriptive LNP cascade 自动当作 anatomy/mechanism。
2. complex-cell STA=0 就说无 stimulus selectivity。
3. 忘记 square 后 sign/phase 信息被改变。
4. 把 task accuracy 与 neural predictivity 当同一指标。
5. surround suppression 等同于 inhibitory firing/conductance 必然增加。
6. 把 ISN 名称理解为 inhibition rate 总是更高。
7. 只看 fixed point 不检查 stability。

Course-specific risk boundary. 明确“复杂细胞 STA=0”不等于“没有 stimulus selectivity”；这是理解 nonlinear system identification 的关键反例。

11 Cumulative Knowledge Check

Do not turn to the Hint Bank until you have written a first attempt.

1. Hubel–Wiesel simple-cell construction 的核心?

2. complex cell STA 为什么可为零?

3. energy model 需要哪两类 filters?

4. 证明 phase invariance 的 identity?

5. task accuracy 高是否保证 neural predictivity 高?

6. model features 与 neural responses 常如何比较?

7. surround suppression 是否要求 inhibition 增加?

8. nullcline intersection 表示什么?

9. surround input 如何改变 fixed point?

10. ISN 的 E subnetwork 特征?

11. steady-state I rate 下降是否排除 ISN?

12. STA 与 STC 分别检测哪类 moments?

13. 公式表中的 “motion/orientation energy model” 解决什么问题? 写出适用条件与一个 sanity check。

14. 公式表中的 “phase-invariance identity” 解决什么问题? 写出适用条件与一个 sanity check。

15. 公式表中的 “model-to-neuron linear mapping” 解决什么问题? 写出适用条件与一个 sanity check。

16. 公式表中的“E/I nullclines” 解决什么问题？ 写出适用条件与一个 sanity check。

12 Hint Bank

1. **Hint 1.** 先识别对象、变量与假设。
Hint 2. 写出最相关的定义或方程，再代入。
Hint 3. 检查单位、shape、符号与极限情况。
2. **Hint 1.** 先识别对象、变量与假设。
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Hint 2. 写出最相关的定义或方程，再代入。
Hint 3. 检查单位、shape、符号与极限情况。
9. **Hint 1.** 先识别对象、变量与假设。
Hint 2. 写出最相关的定义或方程，再代入。
Hint 3. 检查单位、shape、符号与极限情况。
10. **Hint 1.** 先识别对象、变量与假设。
Hint 2. 写出最相关的定义或方程，再代入。
Hint 3. 检查单位、shape、符号与极限情况。
11. **Hint 1.** 先识别对象、变量与假设。
Hint 2. 写出最相关的定义或方程，再代入。
Hint 3. 检查单位、shape、符号与极限情况。
12. **Hint 1.** 先识别对象、变量与假设。
Hint 2. 写出最相关的定义或方程，再代入。
Hint 3. 检查单位、shape、符号与极限情况。
13. **Hint 1.** 先从“motion/orientation energy model”对应的变量关系入手。
Hint 2. 把条件“quadrature filters”逐字转换为 assumption。
Hint 3. 写公式，再检查至少一种极端参数或单位。
14. **Hint 1.** 先从“phase-invariance identity”对应的变量关系入手。
Hint 2. 把条件“ideal pair”逐字转换为 assumption。
Hint 3. 写公式，再检查至少一种极端参数或单位。

15. **Hint 1.** 先从 “model-to-neuron linear mapping” 对应的变量关系入手。
Hint 2. 把条件 “features z fixed” 逐字转换为 assumption。
Hint 3. 写公式, 再检查至少一种极端参数或单位。
16. **Hint 1.** 先从 “E/I nullclines” 对应的变量关系入手。
Hint 2. 把条件 “steady state” 逐字转换为 assumption。
Hint 3. 写公式, 再检查至少一种极端参数或单位。

13 Complete Answer Key with reasoning

1. 按空间排列汇聚多个 center-surround subunits, 形成 oriented ON/OFF lobes。
2. 对相反 phases 都响应, 平均抵消。
3. 同 orientation/frequency、phase 差 $\pi/2$ 的 quadrature pair。
4. $\cos^2\phi + \sin^2\phi = 1$ 。
5. 不保证; 二者需分别测量。
6. 拟合简单 readout (常为 linear mapping) 再 cross-validate。
7. 不要求; excitation 可下降更多。
8. 可能的 steady state。
9. 移动一条或两条 nullclines, 改变 intersection。
10. 在 operating point 单独会不稳定/放大, 由 feedback I 稳定。
11. 不排除, stabilization 是 dynamical feedback property。
12. STA 检测 conditional mean; STC 检测 covariance。
13. 适用条件/约定: quadrature filters. sanity check 应检查单位、符号、极限或 shape 是否与定义一致。

$$E = (K_1 * s)^2 + (K_2 * s)^2$$

14. 适用条件/约定: ideal pair. sanity check 应检查单位、符号、极限或 shape 是否与定义一致。

$$\cos^2 \phi + \sin^2 \phi = 1$$

15. 适用条件/约定: features z fixed. sanity check 应检查单位、符号、极限或 shape 是否与定义一致。

$$\hat{r} = Bz$$

16. 适用条件/约定: steady state. sanity check 应检查单位、符号、极限或 shape 是否与定义一致。

$$\dot{r}_E = 0, \dot{r}_I = 0$$

14 Source-page concordance and coverage audit

Source file	Page	Coverage status
Lecture17-VisualNetworks.pdf	1	Main source-page unit: thumbnail, reconstruction, explanation, prediction question.
Lecture17-VisualNetworks.pdf	2	Main source-page unit: thumbnail, reconstruction, explanation, prediction question.
Lecture17-VisualNetworks.pdf	3	Main source-page unit: thumbnail, reconstruction, explanation, prediction question.
Lecture17-VisualNetworks.pdf	4	Main source-page unit: thumbnail, reconstruction, explanation, prediction question.
Lecture17-VisualNetworks.pdf	5	Main source-page unit: thumbnail, reconstruction, explanation, prediction question.

15 Errata, uncertainty log, and external endnotes

15.1 Errata / source cautions

1. Page 1 的 complex-cell null STA 是关键 nonlinear system-identification 反例。
2. Page 3 的 performance-optimized models 提供 correspondence, 不证明 biological training mechanism。
3. Page 5 明确称 ISN 名称需谨慎理解; 本教材保留这一边界。

15.2 Uncertainty log

No unresolved handwriting uncertainty changes a core definition, equation, figure interpretation, or answer in this companion. Where handwriting was potentially ambiguous, the source-page image is preserved and the reconstruction avoids claiming unverified detail.

15.3 External endnotes

No external web or textbook source was used to replace the uploaded notes. Added material is limited to necessary mathematical scaffolding, explicit assumptions, standard sanity checks, and clearly labeled transfer examples.

15.4 Final self-audit

All source pages listed above were visually inspected. Equations were checked for symbol definitions, units/shapes, assumptions, algebraic transitions, limiting cases, and failure conditions. The PDF is static and contains no interactive form fields or scripts.